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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

**School of Electronics and Communication
Engineering Final Year B. Tech. (ECE)**

**Artificial Intelligence and Machine Learning
Semester-VIII**

Problem Statement 64 - Apply Min-Max Scaling.

By

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Introduction:

Feature scaling is an important preprocessing step in machine learning used to normalize the range of independent variables or features of data. One of the most commonly used normalization techniques is **Min-Max Scaling**.

The main goal of Min-Max Scaling is to transform features into a fixed range, usually between 0 and 1. This ensures that no feature dominates others due to its larger magnitude. It is especially useful in distance-based algorithms like K-Nearest Neighbors (KNN) and clustering algorithms.

This project demonstrates how to implement Min-Max Scaling in Python using NumPy and Scikit-learn and visualize the effect of scaling using Matplotlib.

Objective:

The objectives of this project are:

- To understand the concept of Min-Max Scaling
- To implement Min-Max Scaling using Python
- To transform features into a normalized range [0,1]
- To compare model performance before and after scaling
- To visualize the effect of scaling on data

Dataset Used:

A small dataset is used in this project.

Age	Salary	Purchased
18	20000	0
22	25000	0
25	30000	0
30	40000	1
35	50000	1
40	60000	1
45	65000	1
50	70000	0
55	80000	1
60	90000	1

Algorithms:

Step 1: Find Minimum and Maximum Values

Identify the minimum and maximum values for each feature.

Step 2: Apply Min-Max Formula

Transform each value using the formula:

$$X_{\text{scaled}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

Step 3: Scale the Dataset

Apply the formula to all feature values to bring them into range $[0,1]$.

Step 4: Train Model

Train a machine learning model using both scaled and unscaled data.

Step 5: Compare Results

Evaluate and compare performance before and after scaling.

Result:

Accuracy Comparison

- Accuracy WITHOUT Scaling: (varies based on run)
- Accuracy WITH Min-Max Scaling: Improved accuracy observed

Observations:

- After scaling, all feature values lie between 0 and 1
- Features contribute equally to the model
- Distance-based model performance improves

Visualization:

- Left graph shows original data distribution
- Right graph shows scaled data distribution
- Scaled data is normalized and evenly distributed

Applications of Min-Max Scaling:

Min-Max Scaling is widely used in many domains:

- K-Nearest Neighbors (KNN)
- Neural Networks

- Clustering algorithms (K-Means)
- Image processing
- Data preprocessing pipelines

Conclusion:

This project successfully demonstrates the implementation of Min-Max Scaling for feature normalization. The technique transforms data into a uniform scale, improving the performance of machine learning algorithms. Visualization using Matplotlib helps understand how data distribution changes after scaling. Min-Max Scaling is a simple yet powerful preprocessing technique widely used in real-world applications.